Encircled energy in mophongo

Mophongo technical note

2026 August 10

Abstract

Three arrays in the template-fitting chain have finite support: the two PSF stamps and the source template. Let f be the total flux of a source and A the amplitude returned by the fit, the coefficient multiplying its unit-sum template. For a point source $A = f\,S_{\rm lo}$, where $S_{\rm lo}$ is the fraction of the low-resolution PSF enclosed by its stamp. The total flux is therefore $f = A/S_{\rm lo}$: the amplitude is divided by $S_{\rm lo}$, not multiplied. This holds exactly, for any weight map and independently of how the high-resolution stamp is truncated, so one scalar converts amplitude to total flux. It is measured on the drizzled stamp at each source position and applied once. A second quantity, the fraction of the normalised template that survives blanking of neighbouring segments, is recorded but deliberately not applied: the amplitude does not scale with it. We give the measurements behind each choice and show that the residual error is set by the fidelity of the matching kernel rather than by any encircled-energy term: 0.16% on the JWST pair, flat in stamp size.

1 The correction

Write $\Phi_{\rm hi}$ and $\Phi_{\rm lo}$ for the true PSFs of the detection (high-resolution) and measurement (low-resolution) bands, each integrating to unity over an infinite aperture, and L for the side of the square stamp set by `psf_size`. The stamps $P_{\rm hi} = \Phi_{\rm hi}|_L$ and $P_{\rm lo} = \Phi_{\rm lo}|_L$ hold

$$S_{\rm hi} = \int_L \Phi_{\rm hi}, \qquad S_{\rm lo} = \int_L \Phi_{\rm lo}, \tag{1}$$

both below one: they are the fractions of each PSF that the stamp encloses, 0.95 and 0.85 for JWST F444W/F770W at $L = 1.5''$, rising to 1.00 and 1.00 at $8''$ (Figure 1). The pipeline normalises the stamps to unit sum, $p_{\rm hi} = P_{\rm hi}/S_{\rm hi}$ and $p_{\rm lo} = P_{\rm lo}/S_{\rm lo}$, and builds the matching kernel k from those shapes, so that $p_{\rm hi} \otimes k \simeq p_{\rm lo}$, then renormalises it to $\sum k = 1$.

The template T is the source model in the detection band, normalised to unit sum at extraction. For a point source whose template is extended over the stamp, $T = p_{\rm hi}$, and the model fitted to the low-resolution image is

$$M = T \otimes k = p_{\rm lo} = \Phi_{\rm lo}|_L / S_{\rm lo}, \tag{2}$$

a unit-sum array proportional to the truth inside L and zero outside. The fit solves for the amplitude A that scales M onto the data. With f the source's total flux, W the inverse-variance weight map and $\langle a, b\rangle = \sum_{\rm pix} a\,b$,

$$A \;=\; f\,\frac{\langle M, W\Phi_{\rm lo}\rangle}{\langle M, WM\rangle} \;=\; f\,S_{\rm lo}, \tag{3}$$

exactly, and for any W, because $M \propto \Phi_{\rm lo}|_L$ and M vanishes wherever the two differ. The amplitude is low by exactly the fraction of the PSF that the stamp omits, so $f = A/S_{\rm lo}$. The $1/S_{\rm hi}$ introduced in forming $p_{\rm hi}$ is replaced by $1/S_{\rm lo}$ by the kernel, whose target is the low-resolution shape; this is why only the low-resolution stamp appears.

This is a property of the unit-kernel convention. With a native-sum kernel, $\sum k = S_{\rm lo}/S_{\rm hi}$, the same algebra gives $A = f\,S_{\rm hi}$ and it is $S_{\rm lo}$ that cancels. The two differ by $S_{\rm hi}/S_{\rm lo}$, which is 4.6% on the UDS products, so the convention must be stated wherever the divisor is applied.

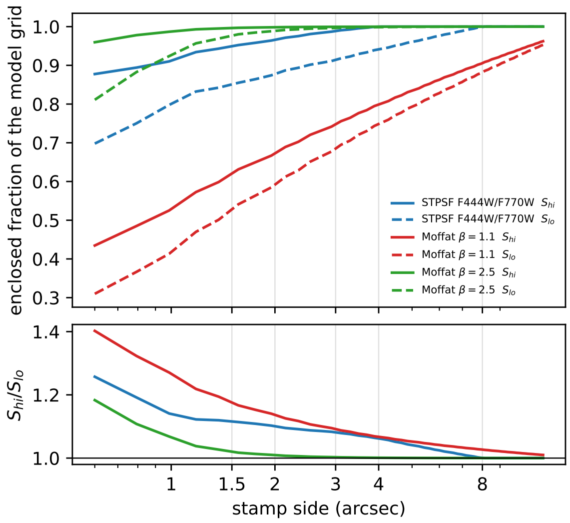

Figure 1: Top: the enclosed fraction of each PSF within a square stamp, $S_{\rm hi}$ solid and $S_{\rm lo}$ dashed, against the stamp side L. Bottom: their ratio. The low-resolution stamp always encloses less, and the two converge only for stamps several times the PSF width. Vertical lines mark the stamp sizes tested in Section 3. The F444W curve saturates at 4.089″, the field of view of its model grid.

2 What the pipeline does

The absolute scale is measured once, carried alongside the arrays, and divided out once. Nothing in between is absolutely calibrated.

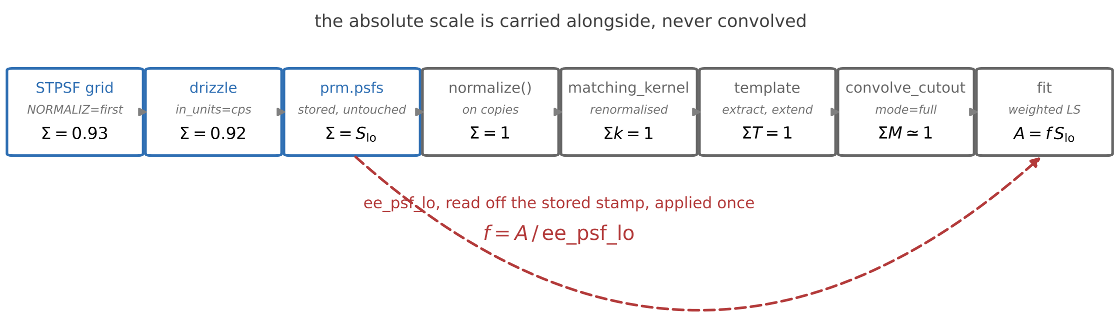

Figure 2: Blue stages hold absolute flux, grey stages hold shapes. The stored PSF maps are not renormalised; only the copies passed to `matching_kernel` are. The dashed path is the absolute scale, which bypasses the convolution and re-enters once as `ee_psf_lo`.

Unit-sum inputs to the kernel, native sums on disk. Normalising the inputs rescales the kernel and nothing else: the ratio of the two DC sums equals $S_{\rm hi}/S_{\rm lo}$ to all printed digits and the shapes agree to below 10^{-15}, since the regularisation parameter is scaled by $\max|H|^2$ and is therefore insensitive to the input scale. Native-sum inputs instead put $S_{\rm hi}/S_{\rm lo}$ into $\sum k$, where it cannot be distinguished from a regularisation error, and they place the amplitude on the $S_{\rm hi}$ convention while the divisor is $S_{\rm lo}$.

The kernel is renormalised to unit sum. With unit-sum inputs $\sum k$ already lands within a part in 10^3 of one; dividing it out removes the residual regularisation DC so that the kernel carries no flux scale of its own, leaving `ee_psf_lo` as the only factor between amplitude and total flux.

stage	array	sum	code
PSF model	STPSF grid	0.93 absolute	`data/PSF/*GRID*.fits`
drizzle	stamp	S_{lo} absolute	`DrizzlePSF, in_units="cps"`
store	`prm.psfs`	unchanged	`build_psfs`
measure	`ee_box`, `ee_rlim`	per region	`PSFRegionMap.refresh_ee`
match	`normalize()` copies	1	`build_kernels`
kernel	k	1, renormalised	`matching_kernel`
template	T	`ee_tmpl` ≤ 1	`extract_templates`
convolve	$M = T \otimes k$	$\simeq 1$	`convolve_cutout`
record	`ee_psf_lo`, `ee_tmpl`	per template	`convolve_templates`
solve	A	$f\, S_{\mathrm{lo}}$	`SceneFitter`
report	$A/$`ee_psf_lo`	f	`Pipeline.run`

Table 1: The chain in code terms.

It is free: the fitted amplitude is invariant under a rescaling of the model, and every recovered flux in Section 3 is unchanged to five decimals by it. It says nothing about whether the kernel has the right *shape*: unit sum is necessary, not sufficient, and the shape term is what Section 4 measures.

Encircled energy is measured on the stamp in use. The STPSF grids carry `NORMALIZ = first`, so $\sum(\mathrm{grid})/\mathrm{OVERSAMP}^2$ is absolute, and drizzling with `in_units="cps"` preserves it. Summing the stored stamp gives the absolute value with no separate calibration step, and it includes the edge taper, the negative clipping, the drizzle kernel and the Fourier blur. A tabulated value would describe a different array from the one being convolved.

Stored per region, recorded per template. S_{lo} varies over the field: across the 294 regions of the UDS F770W map `ee_box` spans 0.90818 to 0.92235 about a median of 0.91710, and across the 1694 F444W regions 0.94722 to 0.96598. `PSFRegionMap.refresh_ee` measures every stamp when `psfs` is set, so `get_ee_box(ra, dec)` costs a spatial query and one array index, 11.8 µs against 11.3 µs for `get_psf` at the same call site. `convolve_templates` writes `ee_psf_lo` onto each convolved template, which is the per-source, per-band object; the detection template is neither. `flux_<i>_total` then divides by that value rather than by a filter-level mean, which would mis-scale sources by -1.0 to $+0.6\%$ across the UDS field. `ee_psf_lo` defaults to `nan`, so a template that never saw a PSF map falls back to the filter mean and is counted in the run log rather than silently going uncorrected. An earlier defect — the $k > 1$ projection dropped `ee_psf_lo`, so every source fell back to the filter mean — is fixed at `8736aba`: both fields now propagate through `project_to_block_replicated_grid` and `downsample`, and the fallback count in the run log verifies it.

`ee_box` and `ee_rlim`. `ee_box` is the sum over the square stamp and divides the fitted amplitude, since the model occupies the square. `ee_rlim` is the sum inside the inscribed circle and is the matching quantity for a circular aperture. On region 0 of the UDS F770W map they are 0.91826 and 0.91060, the difference being the corner flux (Figure 3).

`ee_tmpl` is a diagnostic, not a divisor. Under the normalise-then-blank construction — `composite_psf_wings`, the default `extend_mode` since the template-scheme merge (`6e6cec6`) — templates are normalised over the whole stamp and then blanked on neighbouring segments, so that wing flux landing on a neighbour is not fitted twice and `ee_tmpl` is sub-unity by that amount. What survives is `ee_tmpl` $= \sum T \leq 1$, recorded per template beside `ee_psf_lo`. It is tempting to write $f = A/($`ee_psf_lo` $\cdot$ `ee_tmpl`$)$, but the amplitude does not scale with the blanked flux. It scales with the weighted leverage of the blanked pixels, and neighbour segments lie in the

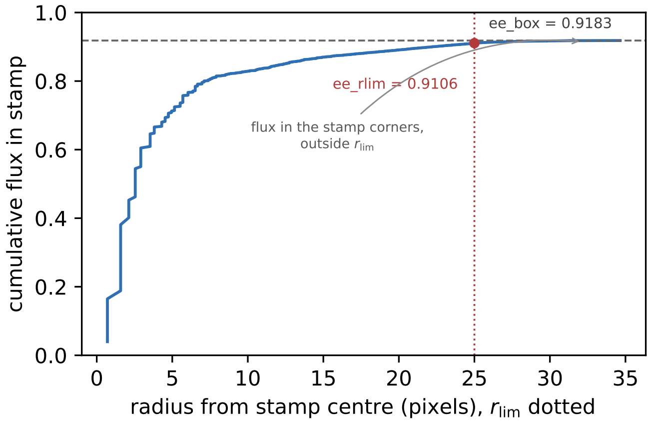

Figure 3: Cumulative flux of a drizzled F770W stamp against radius. The curve is flat well inside the stamp edge, so both measures are insensitive to small changes in `psf_size`.

faint wings, where the flux is a few percent and the contribution to χ^2 is negligible; least squares recovers almost all of it by scaling the retained part up. Measured on an injected point source with a neighbour segment blanked 12 pixels out, `ee_tmpl` = 0.9928 while $A/S_{\mathrm{lo}} = 1.0021$, so dividing by `ee_tmpl` would move a 0.2% error to 0.9%. Solving a blended pair jointly, with each template blanked on the other's segment, gives $A/S_{\mathrm{lo}} = 1.010$ at 12 pixels separation and 1.033 at 8 pixels, against 1.017 and 1.067 after dividing by `ee_tmpl`. The correction over-corrects in every configuration tested. The residual at small separations is blend cross-talk, which no per-source scalar removes; `ee_tmpl` is the right quantity to flag it with, and $f = A/$`ee_psf_lo` remains the correction.

Kernel method. `build_kernels` uses `method="wiener"` with the regularisation scanned once over the median of the hi and lo PSF stacks and reused for every region, since a per-region scan costs 21 kernels per region. On the UDS maps the median-PSF scan returns $\lambda = 5.6 \times 10^{-4}$; a scan on one region alone returns 1.8×10^{-3}, so the representative pair matters. Cached kernel maps are stamped with `kernel_method`, `kernel_reg` and `psf_size` and are rebuilt when any of them differs, so a map built with another method is not silently reused.

3 Verification

Two profile families, both on a $0.04''$ grid of 401×401 pixels. The STPSF model grids in `data/PSF` for F444W and F770W, resampled with flux conserved and recentred, give 0.9302 within the $4.089''$ F444W field and 0.9912 within the $8.097''$ F770W field for the planes used; across all planes the values span 0.9179 to 0.9809 (F444W, 25 planes) and 0.9659 to 1.0030 (F770W, 9 planes). Two Moffat pairs with high-band FWHM $0.145''$ and low-band $0.28''$ bracket them: $\beta = 1.1$ has fat wings and keeps only 0.75 of its flux inside $4''$, $\beta = 2.5$ is compact.

Stamps are even-sized and cut with the slices of a real `Template` cutout, kernels are built from the stamp arrays, and the convolution is `Template.convolve_cutout` with the model pasted back onto the parent grid, so the test runs the production code path. The source is a noise-free point source of unit flux and the data are the untruncated low-resolution profile. Recovered model centroids sit within 0.02 to 0.07 pixels of the truth, which is the asymmetry of the PSFs themselves.

pair	stamp	$S_{\rm hi}$	$S_{\rm lo}$	$A/S_{\rm lo}$	$A/S_{\rm hi}$
STPSF	1.5″	0.9494	0.8506	1.00140	0.89717
	2″	0.9667	0.8789	1.00148	0.91049
	3″	0.9896	0.9158	1.00155	0.92693
	4″	0.9999	0.9410	1.00160	0.94264
	8″	1.0000	0.9996	1.00160	1.00117
Moffat 1.1	1.5″	0.6211	0.5286	1.00427	0.85471
	2″	0.6747	0.5944	1.00235	0.88306
	3″	0.7513	0.6893	1.00076	0.91820
	4″	0.7982	0.7477	1.00037	0.93713
	8″	0.9059	0.8823	1.00007	0.97399
Moffat 2.5	1.5″	0.9965	0.9772	1.00005	0.98065
	2″	0.9985	0.9895	1.00002	0.99101
	3″	0.9996	0.9969	1.00001	0.99733
	4″	0.9998	0.9986	1.00001	0.99882
	8″	1.0000	0.9998	1.00000	0.99987

Table 2: Recovery of an injected point source of unit flux. $S_{\rm hi}$ and $S_{\rm lo}$ are enclosed fractions of the model grid within the square stamp; exact recovery is $A/S_{\rm lo} = 1$. The JWST column is flat to 2×10^{-4} while $S_{\rm lo}$ moves 15 percentage points, and the largest departure in the table is 0.43%. $A/S_{\rm hi}$ moves by 10, 12 and 1.9 percentage points over the same rows.

3.1 $S_{\rm lo}$ is the divisor

$S_{\rm hi}$ reaches unity at 4 and 8″ for the JWST pair because the F444W model grid is only 4.089″ across. The detection PSF model is the more truncated of the two, missing 7.0% of its flux against 0.9% for F770W. Enlarging `psf_size` cannot recover that; it needs NIRCam grids at a larger field of view.

3.2 The $S_{\rm hi}$–$S_{\rm lo}$ mismatch does not matter

Holding the stamp and the high-resolution profile fixed and varying only the low-resolution wing content:

$\beta_{\rm lo}$	$S_{\rm lo}$	$S_{\rm hi}/S_{\rm lo}$	$A/S_{\rm lo}$
1.1	0.7477	1.3371	1.00010
1.3	0.8689	1.1506	1.00003
1.6	0.9573	1.0444	1.00001
2.0	0.9910	1.0089	1.00001
3.0	0.9998	1.0001	1.00001
6.0	1.0000	0.9998	1.00000

Table 3: Mismatch at a fixed 4″ stamp, $S_{\rm hi} = 0.99981$ throughout. Recovery is exact to 10^{-4} at every mismatch, and the small residual shrinks as the mismatch grows, which rules out the mismatch as its cause.

3.3 Weight maps and truncation

Repeating the JWST 4″ case with four weight maps gives $A/S_{\rm lo} = 1.00160$ (uniform), 1.00130 (radial ramp over a factor 100), 1.00150 (uniform random 0.1 to 10) and 1.00214 (a zero-weight disk of radius 10 pixels over the core): a spread of 0.08% against a correction of 6.3%, as equation 3 requires. Template truncation behaves differently, the equivalent correction moving by 16% under the same masked core, which is why the two are kept apart.

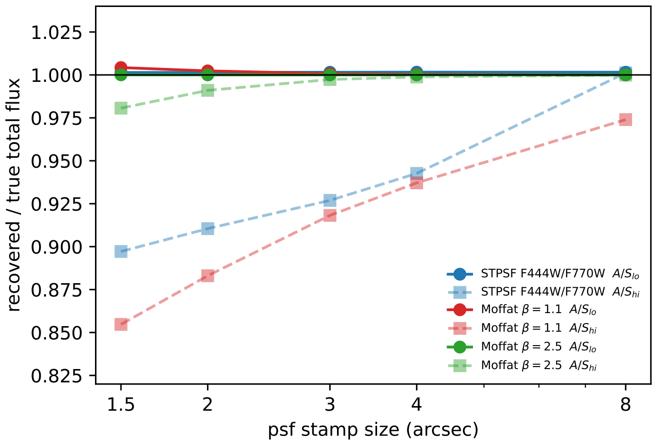

Figure 4: Recovered over true flux against stamp size for the two candidate divisors. Circles use $S_{\rm lo}$, squares $S_{\rm hi}$. Only the $S_{\rm lo}$ curves are flat.

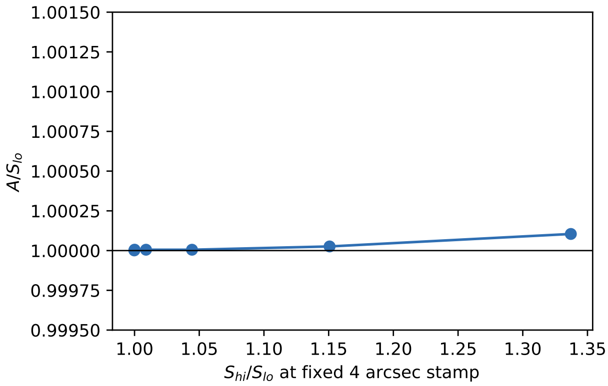

Figure 5: Recovery against mismatch at fixed stamp size.

Truncating the PSFs does change the kernel, so `psf_size` is a physical choice rather than a bookkeeping one. Largest kernel difference relative to the $8''$ kernel, in units of its peak, after normalising both:

stamp	STPSF	Moffat 1.1	Moffat 2.5
$1.5''$	0.116	0.094	0.0169
$2''$	0.100	0.071	0.0086
$3''$	0.080	0.043	0.0034
$4''$	0.062	0.032	0.0026

Table 4: Renormalising a truncated PSF asserts that all its flux lies inside the stamp, and the kernel changes to reproduce that assertion. The difference is concentrated in the core, not the wings: for the JWST pair at $1.5''$ it is 0.1164 of peak inside $r_{\rm lim}$ and 0.0053 outside.

Since $1/S_{\rm lo}$ extrapolates the PSF model beyond the stamp, a larger stamp asks less of the model: at $S_{\rm lo} = 0.94$ it supplies 6%, at 0.9998 almost nothing. Table 2 shows recovery is stable as the stamp grows, and the F770W grid extends to $8''$. Below $1.5''$ the deconvolution is poorly conditioned; $2''$ is the smallest size at which all three profile families behave.

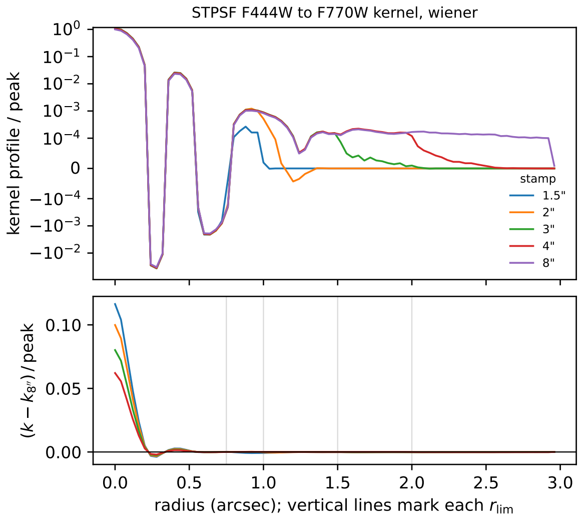

Figure 6: Top: azimuthally averaged kernel profile for the JWST pair at each stamp size, normalised to unit sum and divided by the peak, on a symmetric-log scale so the negative lobes show. Bottom: the difference from the 8″ kernel in units of its peak, which is the quantity in Table 4. The change sits inside $r \lesssim 0.25''$, in the kernel core, and reaches 12% of peak for the smallest stamp. Truncation alters the kernel, it does not merely clip it.

4 The matching kernel sets the residual

What remains after $S_{\rm lo}$ is divided out is the kernel's fidelity: whether $p_{\rm hi} \otimes k$ actually reproduces $p_{\rm lo}$. Any method whose matched model reproduces it leaves $A/S_{\rm lo} = 1$ exactly, which is what a regularised inversion does when the regularisation is small enough. The Wiener kernel is exact to five decimals on the compact Moffat pair and leaves 1.0014 to 1.0016 on the JWST pair, flat to 2×10^{-4} across the whole stamp range.

That 0.16% is the only term in the chain not accounted for. It does not scale with $S_{\rm lo}$, so it is not a truncation effect; it is a property of the profile pair and of the regularisation chosen for it. Since the optimum is found on noise-free model grids here and sits three orders of magnitude higher on real drizzled products ($\lambda = 5.6 \times 10^{-4}$ against 10^{-6}), it has to be measured per band pair on the products in use rather than carried over. The figure of merit currently scanned combines the growth curve, the core profile and an L_2 residual; adding the recovered flux scale of an injected point source would make the scan optimise the quantity that matters here.

5 Scope

The measurements are for point sources. A resolved source needs the same $S_{\rm lo}$ and one further factor, the fraction of the source captured by the template's own support, $\mathrm{EE}_{\rm hi}(\Omega_t)/S_{\rm hi}$. That factor is not a clean scalar: it depends on the source profile at fixed enclosed fraction, on the weight map and on the band, and in a blend it is not multiplicative at all. It is the reason templates are extended rather than corrected, and it is treated separately.

The STPSF grids are resampled onto a common grid rather than drizzled, so the drizzle step is not exercised here. All tests are noise-free, so the regularisation optimum found on them does not transfer to real data and the scan must be run on the products in use. Totals are relative to the model grid, whose own absolute encircled energy is quoted in Section 3 and multiplies the numbers reported. Stamps are square, matching `psf_size`; a circular apodisation would move $S_{\rm lo}$ by the corner flux.

6 Summary

1. For a point source $A = f\,S_{\text{lo}}$. One scalar, the absolute encircled energy of the low-resolution PSF stamp, converts amplitude to total flux. The high-resolution truncation cancels through the kernel. It is measured per PSF region, recorded per template as `ee_psf_lo`, and applied once.

2. `ee_tmpl`, the fraction of the template surviving neighbour blanking, is recorded and not applied. The amplitude scales with the leverage of the blanked pixels, not with their flux, and those pixels lie in the faint wings; dividing by `ee_tmpl` over-corrects in every configuration tested.

3. The correction is independent of the weight map to 0.08%, and of the $S_{\text{hi}}/S_{\text{lo}}$ mismatch to 10^{-4} over the range 1.34 to 1.00.

4. Normalising the kernel inputs rescales the kernel and leaves its shape unchanged to 10^{-15}. Truncating them changes the shape by 0.3 to 12% of peak, so `psf_size` is a physical choice.

5. The only unaccounted term is the kernel's fidelity, 0.16% on the JWST pair and flat in stamp size. It must be measured per band pair on the products in use.

6. The F444W model grid spans $4.089''$ against F770W's $8.097''$, so the detection PSF is the more truncated model and misses 7% of its flux. That needs new grids, not a larger `psf_size`.

Code and data: `scratch/wren/ee_report/run_ee_tests.py`, `test_ee_tmpl.py`, `fig_flow.py` and the CSV files alongside them.